

DevOps Engineering Foundation - 6-Week Intensive Bootcamp

Course Overview

This comprehensive curriculum transforms absolute beginners into job-ready DevOps engineers through hands-on projects, curated resources, and real-world scenarios. By Week 6, you'll have a production-grade portfolio project demonstrating CI/CD, containerization, orchestration, infrastructure as code, and monitoring.

WEEK 3: Containerization with Docker

Learning Outcomes:

- Understand container concepts and Docker architecture
- Create Dockerfiles and build custom images
- Manage multi-container applications with Docker Compose
- Push images to Docker Hub and container registries

Resources:

Docker Fundamentals:

- [Docker Tutorial for Beginners - Full Course](#) by freeCodeCamp (3 hours)
- [Docker Crash Course for Absolute Beginners](#) by TechWorld with Nana (1 hour)
- [Docker Official Documentation - Get Started](#)
- [Play with Docker](#) - Interactive hands-on playground

Docker Compose:

- [Docker Compose Tutorial](#) by TechWorld with Nana (30 min)
- [Docker Compose Documentation](#)

Best Practices:

- [Docker Best Practices for Developers](#) by Docker (45 min)
- [Dockerfile Best Practices](#)

Week 3 Project:

Containerized Full-Stack Application

Build and containerize a complete application stack:

1. Find an app built with (Frontend, Backend, Database)
 - Frontend: React/Vue.js or simple Nginx serving static content
 - Backend: Node.js/Python Flask REST API
 - Database: PostgreSQL or MongoDB
2. Write Dockerfiles for each component
3. Create a docker-compose.yml to orchestrate all services
4. Implement environment variables for configuration
5. Set up persistent volumes for database data
6. Configure proper networking between containers
7. Push all images to Docker Hub
8. Create comprehensive documentation

Deliverables:

- GitHub repository with all Dockerfiles and docker-compose.yml
- Docker Hub links to all images
- README with architecture explanation and setup instructions
- Environment variables template file
- Screenshot/video of running application
- Performance comparison (container vs. non-container)